The Effect of Magnetic Saturation on Sensorless Control of a Brushless Permanent Magnet Motor under AC and DC Excitation

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***Abstract* – The design and performance of a permanent magnet rotor for a concentrated wound, torque dense machine exhibiting a saliency profile suited to high-frequency injection based zero- and low-speed sensorless control is presented. Rotor design is mechanically constrained by an existing stator. Electromagnetic specifications require the motor to match the torque capability of an existing baseline machine whilst demonstrating a saliency characteristic suited for injection based, low-speed sensorless control. The effects of fundamental current amplitude on flux saturation and saliency profile are investigated for brushless DC and brushless AC operation. The saliency profiles required for sensorless detection show significant differences under these operating modes. For the selected rotor design, brushless AC operation results in a stronger saliency signature at light loads, which quickly deteriorates with the onset of saturation. Brushless DC operation results in more gradual deterioration in saliency signature, suggesting sensorless control would be possible over a greater torque range.**

***Index Terms* – PM motor, sensorless control, AC and DC operation, design**

I. Introduction

Brushless permanent magnet (PM) machines offer a power dense actuation technology that is increasingly being deployed in "more-electric" applications in the aerospace and automotive sectors [1]. Correct commutation of the machine phases requires knowledge of the angular position of the permanent magnet rotor which, for brushless DC operation, is usually achieved using Hall-effect transducers [2]. For high-performance brushless AC operation, a resolver or encoder is typically required which may add considerable cost to the drive system whilst reducing its ruggedness. Sensorless control methods, where the rotor position is estimated from measurements of the machine's voltages and currents, can be applied to overcome this disadvantage [3,4]. Sensorless position estimation relies upon the existence of a measurable electrical parameter that is related to rotor angular position. The back-emf can be used when the machine is operating at a sufficiently high speed [5], but at low- and zero-speed methods that typically depend upon detection of the machine's rotor dependent inductance profile are used [6,7]. A permanent magnet machine may exhibit some degree of magnetic saliency, resulting in a rotor position dependent variation in stator winding inductance that can be detected by the injection of a low-magnitude, high-frequency voltage set and measurement of the resulting currents [8]. It can be demonstrated that a signal ε containing information relating to the rotor angle estimation error θ_{err} can be detected from the high-frequency components of the stator currents when a high-frequency voltage vector, $V_i \sin(\omega_i t)$, is superimposed onto the fundamental excitation. Assuming negligible cross-coupling between the d- and q-axes, the signal ε is given by (1) [9]

$$\varepsilon = k \frac{V_i}{\omega_i \, L_q} (\chi - 1) \sin(2\theta_{err}) \tag{1}$$

where L_q is the q-axis inductance and $\chi = L_q / L_d$ is the saliency ratio. When a rotating injected voltage vector is used, $k = -1$, whilst for a pulsating voltage injected into the estimated d-axis, $k = 1/2$. It is apparent from (1) that $\chi \neq 1$ is a prerequisite for high-frequency injection based sensorless control. In addition, a higher saliency ratio improves sensorless estimation by making the rotor position tracking observer less prone to noise. Machine inductances, and consequently the saliency ratio, are highly dependent upon machine design [10]. The rotor topology influences the choice of injection based sensorless control strategy, particularly under high-load conditions, and rotor geometry influences significantly the accuracy of the position estimation method [11, 12]. For a given machine topology, operating conditions can significantly affect its response to

high-frequency signal injection by altering the magnetic saliency as a consequence of saturation of the flux linkages. Saturation, in turn, depends upon the amplitude of the fundamental current excitation and its distribution in the stator windings. To capitalise on the advantages of sensorless control, therefore, consideration may be given to the saliency profile at the design stage of the permanent magnet machine. Realising the benefits of a torque dense, simple to manufacture and rugged machine, whilst achieving a saliency profile that enables reliable rotor position detection with low torque ripple, presents a significant design challenge. The paper, through finite element (FE) analyses of the flux paths coupled with experimental validations, will demonstrate the influence of saturation on signal injection based sensorless control for a number of alternative rotor designs that meet the required performance specifications. Additionally, the fundamental excitation pattern will be investigated by comparing sinusoidal AC operation with DC operation with rectangular current commutation, demonstrating for the latter an extended torque range suitable for sensorless operation.

II. Machine Design

A design methodology has been developed for permanent magnet machines that directly combines a sensorless position estimation algorithm with FE analysis [13]. For the specific machine used to illustrate the procedure, the design process was further constrained by the requirements for the rotor to fit within an existing stator (comprising a concentrated wound stator with 1.5 slots per pole), to maintain a predefined airgap and to use pre-specified materials whilst meeting stringent aerospace design standards. Magnetic saliency and torque, with the application demanding high peak-to-average torque ratio and low operating duty, are key design parameters. These constraints limit the design variables to the dimensions of the permanent magnet segments, as shown by the proposed interior PM arrangement of Fig. 1. The FE method is used to calculate torque and the magnetic flux linkage of the phase windings at discrete rotor positions over an electrical cycle. The machine's radial flux pattern simplifies the analysis by enabling the use of two-dimensional (2D) FE modelling techniques with the calculation area being limited to the machine's cross section. For simplicity, end effects have been neglected [13]. Due to periodicity, the analysis only requires consideration of two pole-pitches. To simulate rotation of the rotor, the moving band technique is used, and the solver used in the FE analysis considers the magnetostatic field and accounts for material non-linearity. Torque is calculated using the co-energy method and is averaged over successive $60°_{elec}$ commutations of a (0, +1, –1) I_{rated} winding current pattern to emulate brushless DC operation. Saliency torque is accounted for by adjusting the commutation angle to yield maximum average torque. The saliency ratio χ is calculated based on the magnetic flux linkage for a stator phase winding, obtained using (2) which defines the flux linkage in a single coil,

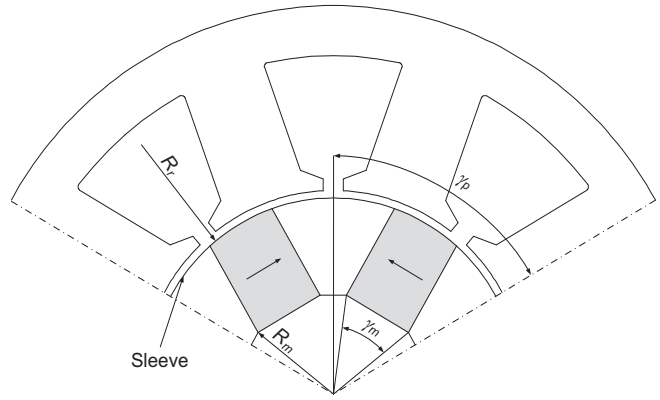

Fig.1. Proposed interior PM arrangement and variables used in the design procedure

where A is the magnetic vector potential (i.e. $A = A_z$), n_t is the number of turns in a single coil, l is the active length of the machine and S is the cross sectional area of a coil for positive or negative current flow only ($S = S_+ = S_-$).

$$\psi = \frac{n_t l}{S}\left(\iint_{S_+} A dS - \iint_{S_-} A dS\right) \tag{2}$$

a)

b)

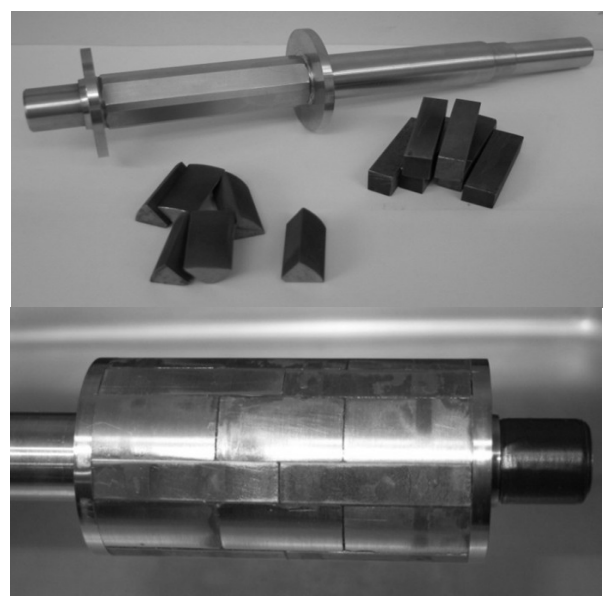

Fig. 2. Interior PM rotor selected for manufacture; a) rotor components; b) assembled rotor (without containment sleeve)

The design process considered 14 possible rotor configurations [13-15] that met both the torque and saliency requirements. Minimization of the error in the estimated position and of the degree of cogging torque introduced by the salient rotor, led to the selection of the best candidate rotor design which was subsequently manufactured, as shown in Fig. 2 [13].

Ideally, in order to assess the suitability of a rotor design for zero- and low-speed sensorless position estimation, an accurate machine model would be combined with the voltage injection system and sensorless algorithm, but the computational complexity of modelling the injection scheme prohibits the use of this direct approach. It has been shown, however, that the behaviour of the sensorless scheme can be accurately reproduced using FE techniques by incorporating a suitable 3-phase current pattern and calculating the resulting flux linkage in the machine winding [14]. For each of the 14 candidate rotor designs, FE simulations were carried out where the machine windings were energised using a balanced 3-phase set of sinusoidal currents superimposed upon the fundamental excitation current. The 3-phase current set is defined by $[I_a\ I_b\ I_c]^T = I_{inj}[\sin(\theta_i)\ \sin(\theta_i\text{-}2\pi/3)\ \sin(\theta_i\text{+}2\pi/3)]^T$, where θ_i is the angle of the injection signal for a given sample. Using the FE post-processor the corresponding flux linkages, ψ_a, ψ_b and ψ_c, of the stator phase coils are found from the vector potential solution for each rotor angle. The computational overhead is minimised by considering the symmetry of the results which enables calculation to be carried out over only one-sixth of an electrical cycle. Flux linkage attributable to the 3-phase current set only, without any contribution from the fundamental excitation current or permanent magnets, is found using $\psi(\theta_i)\big|_{\Theta=const} = \psi_{inj}(\theta_i)\big|_{\Theta=const} - \psi_m(\theta_i)\big|_{\Theta=const}$, where ψ_{inj} is the phase flux linkage including the injected 3-phase set and ψ_m is the flux linkage without signal injection.

III. SALIENCY ANALYSIS

Magnetic saliency is highly dependent upon the operating conditions of the machine. It is well known that fundamental currents of sufficiently high amplitude may drive the machine into saturation affecting sensorless position estimation [10,11]. The current commutation pattern also influences saturation: for example, current may be phase-advanced in order to optimise torque production in brushless DC machines, with the result that the degree of saliency will vary. This, in turn, will affect the accuracy of saliency dependent injection based sensorless rotor position estimation techniques. Simulation results for the variation of saliency ratio as a function of per-unit torque for all of the candidate rotor designs that met the design specification are shown in Figures 3 and 4 for brushless DC operation and brushless AC operation respectively, where a maximum torque per Ampere control strategy is assumed. To assist with interpretation of these results, the plane χ=1 is also plotted in each figure. The degree of saliency required for injection based sensorless position estimation is largely dependent upon the detection scheme. As long as saliency ratio χ>1, injection based sensorless control is theoretically feasible, although it is reasonable to expect that the higher the saliency ratio, the stronger the position dependent measured signal is and the less prone to disturbances the sensorless estimation will be. Figure 3 shows that, for the best candidate rotor (rotor version 3), the machine exhibits saliency χ>1 up to approximately 1p.u. torque. Beyond this torque level, the saliency ratio inverts, i.e. L_q<L_d. The initial saliency is in the region χ=1.7, but decreases as current increases. This is expected as a consequence of saturation. With DC excitation, sensorless control is possible over a wide torque range. Figure 4 shows that with the same machine operated under brushless AC excitation, the maximum saliency ratio χ=2.2 is greater than in the brushless DC case of Figure 3. As current increases, saliency reduces much more rapidly than in the case of DC excitation and the maximum torque achievable under sensorless control (i.e. χ>1) is approximately 0.6p.u. The variations in the behaviour of the machine under different fundamental excitations may be explained by considering Figures 5 and 6, which show the magnetic flux density distributions inside the machine at rated torque, under DC and AC operation respectively. The relative positions of the rotor and stator correspond to the peak torque. It is apparent that the increased peak current required to achieve rated torque under AC operating conditions results in more extended saturated regions (encircled in the figures).

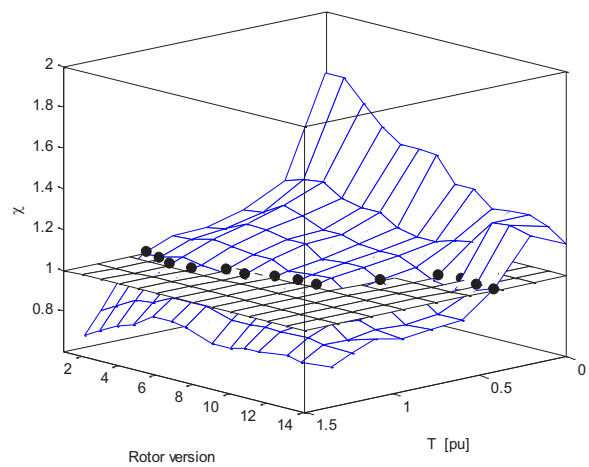

Fig. 3. Variation of the saliency ratio versus rotor version and per unit torque for DC maximum torque per Ampere operation

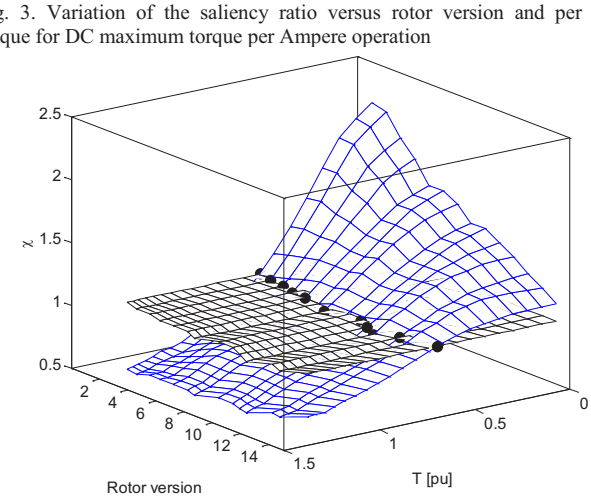

Fig. 4. Variation of the saliency ratio versus rotor version and per unit torque for AC maximum torque per Ampere operation

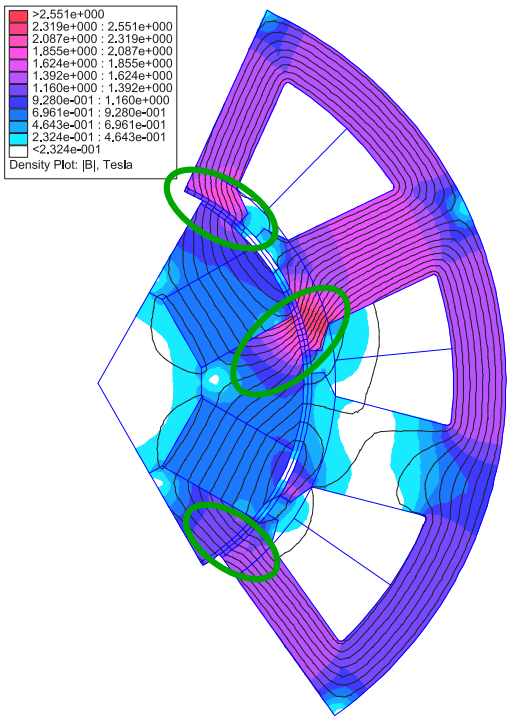

Fig. 5. Flux density for DC operation (28Nm, 98Arms, 120Apk)

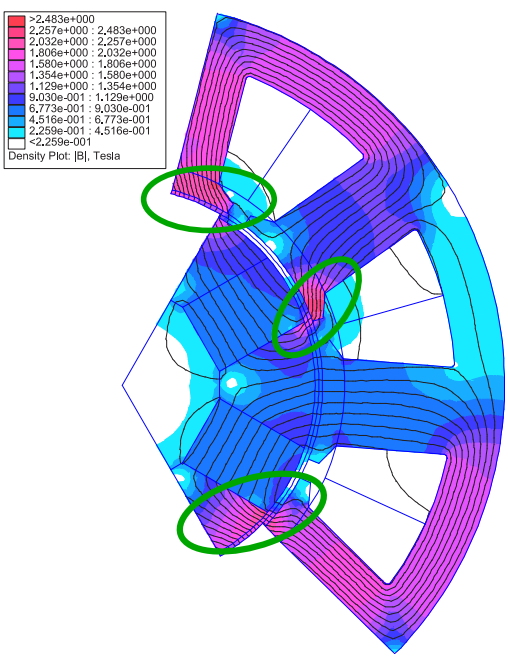

Fig. 6. Flux density for AC operation (28Nm, 98Arms, 138.6Apk)

Although the saliency ratio has been assumed as an important design parameter for sensorless operation, actual performance of the drive will also be affected significantly by cross-saturation which determines a cross-coupling effect between the d- and q- rotor axis.

IV. SENSORLESS ANGLE ESTIMATION

A balanced three-phase high-frequency signal of amplitude V and frequency ω_{hf} is assumed to be injected in the machine stator windings. It results in a rotating vector when observed from the stationary reference frame given by:

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = V \begin{bmatrix} -\cos\omega_{hf}t \\ \sin\omega_{hf}t \end{bmatrix} \tag{2}$$

Assuming negligible cross-coupling, the resulting stator currents in the stationary reference frame are given by:

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = I_1 \begin{bmatrix} \sin\omega_{hf}t \\ \cos\omega_{hf}t \end{bmatrix} - I_2 \begin{bmatrix} \sin(2\theta - \omega_{hf}t) \\ \cos(2\theta - \omega_{hf}t) \end{bmatrix} \tag{3}$$

where the magnitudes of the positive- and negative-sequence components are given by:

$$I_1 = \frac{V}{\omega_{hf}} \frac{L_\Sigma}{L_\Sigma^2 - L_\Delta^2};\ I_2 = \frac{V}{\omega_{hf}} \frac{L_\Delta}{L_\Sigma^2 - L_\Delta^2} \tag{4}$$

and the average and differential inductances by:

$$L_\Sigma = \frac{L_d + L_q}{2}; L_\Delta = \frac{L_d - L_q}{2}$$

It is evident from (3) that the information on the rotor angle θ is contained in the negative sequence component and can be extracted by the simple demodulation technique depicted in Fig. 7 [16].

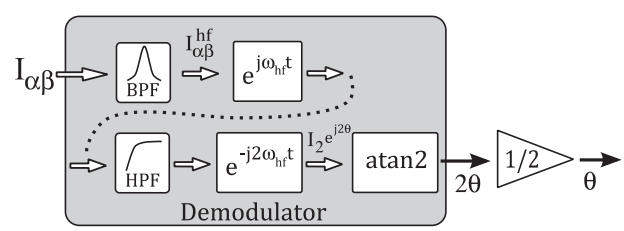

Fig. 7. Demodulation technique

The model in eqs. (2)-(4) and the demodulation technique of Fig. 7 only take into account a single saliency. Additional saliencies due to saturation, asymmetries and slotting effects will generate periodic estimation errors. A number of techniques have however been proposed for the decoupling of higher order saliencies. The input to the four-quadrant inverse tangent function in the demodulator of Fig. 7 is proportional to the negative-sequence current I_2 which, in turn, is proportional to $1/L_q\,(\chi - 1)$, demonstrating how machine saturation and consequent decrease of the q-axis inductance and saliency ratio influence angle estimation.

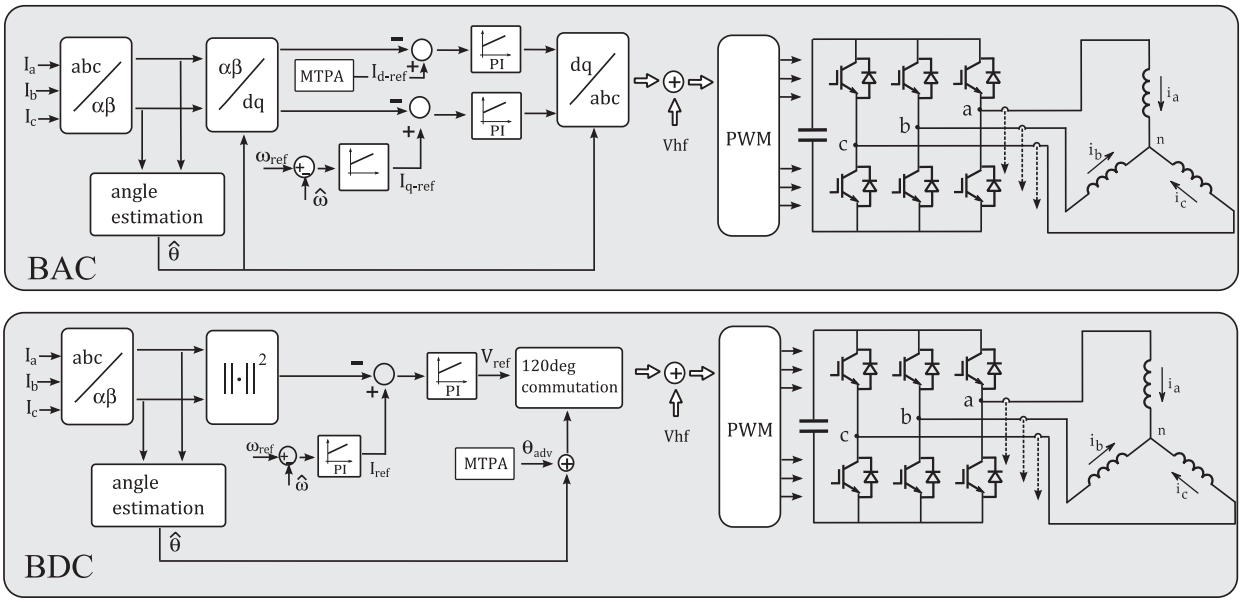

Fig. 9. Schematic of the brushless AC (top) and brushless DC (bottom) motor drive control

V. Experimental Results

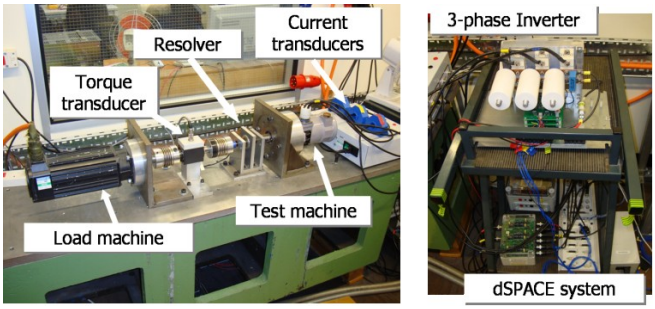

Fig. 8. Experimental test rig showing dynamometer assembly, 3-phase inverter and control platform.

The experimental test rig used for the characterization of the machine is shown in Figure 8. The rig consists of a fully instrumented dynamometer where the brushless permanent magnet test machine is driven using a custom-built 3-phase power MOSFET inverter with 20kHz switching frequency. The dSPACE 1103 control platform is used for the execution of the control algorithm and for data acquisition. The test motor is coupled to a load machine controlled using an industrial drive.

The schematics of the drive control systems for brushless AC and DC operation are shown in Fig. 9. A standard field-oriented control for brushless AC operation is employed, whereas square wave operation with a 120° conduction period is used for brushless DC operation. The same angle estimation algorithm of Fig. 7 is used in both cases, with a 600Hz, 2V carrier superimposed onto the fundamental excitation.

A. Torque and EMF measurements

The FEA based design methodology was verified experimentally by comparing the calculated and measured torques generated by the machine containing the best candidate rotor topology, as shown in Figure 10 Additionally, calculated and measured back-emf waveforms were compared as shown in Figure 11. Both figures highlight the good agreement between the theoretical and practical results and demonstrate the accuracy of the FE design technique.

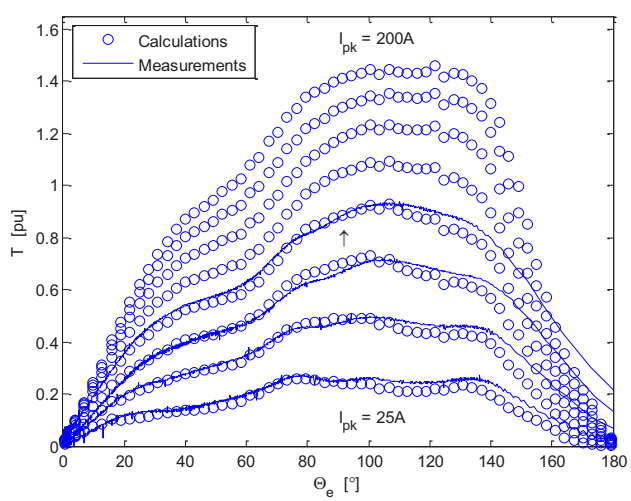

Fig. 10. Angular variation of per unit torque for the PM machine incorporating the best candidate rotor design. Two phases are energised to emulate brushless DC operation.

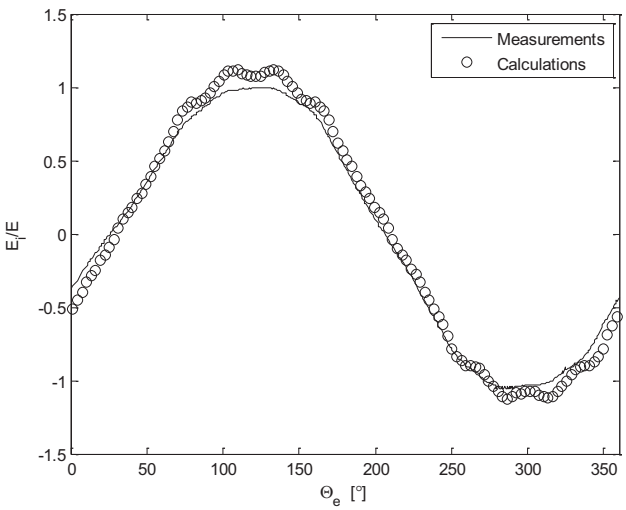

Fig. 11. Normalised line-to-line back-emf for the PM machine incorporating the best candidate rotor design

B. *Angle estimation*

The performances of the angle estimation algorithm have been tested in a number of different operating conditions in order to understand the influence of saturation due to increase in fundamental current under both brushless AC and DC operation. Figs. 12 and 13 show the demodulated $i_{\alpha\beta}$ negative-sequence carrier currents (i.e. the input to the atan function in Fig. 7) and their frequency spectra, respectively, for brushless AC operation with I=0.2-0.8 pu at 10rpm. The decrease in L_q for increased fundamental current compensates the decrease in the saliency ratio and causes an increase in the negative-sequence carrier currents, as it can be explained from (1) and (4). Distortion due to cross-saturation and secondary saliencies, which make the negative sequence carrier currents deviate from ideal sinusoidal waveforms as it is evident from Fig. 12 and the harmonic content shown in Fig. 13, also increases with increased fundamental current. Similar results are reported in Fig. 14 and 15 for brushless DC operation. In this case the increase of the negative-sequence carrier currents is more pronounced making DC operation preferable at higher fundamental currents. These results are confirmed by the performance of the angle estimation algorithm reported in Figs. 16-19. The estimation error in the brushless AC case is significantly larger than the error with brushless DC excitation, making sensorless operation above 0.5pu unfeasible with AC excitation. Sensorless operation with DC excitation is, however, possible over the entire torque range.

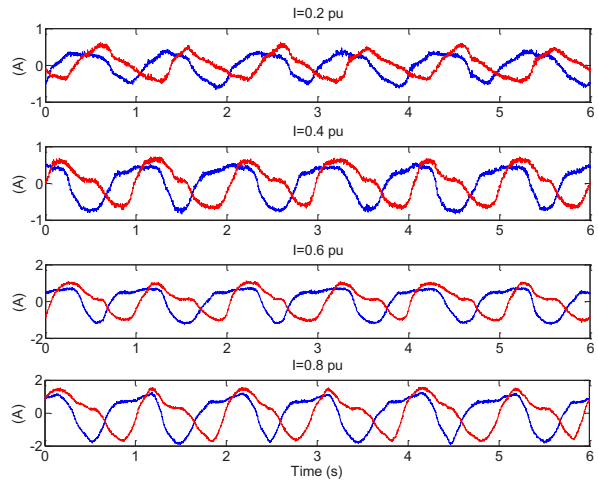

Fig. 12. Demodulated $i_{\alpha\beta}$ negative-sequence carrier currents for brushless AC operation with I=0.2-0.8 pu at 10rpm

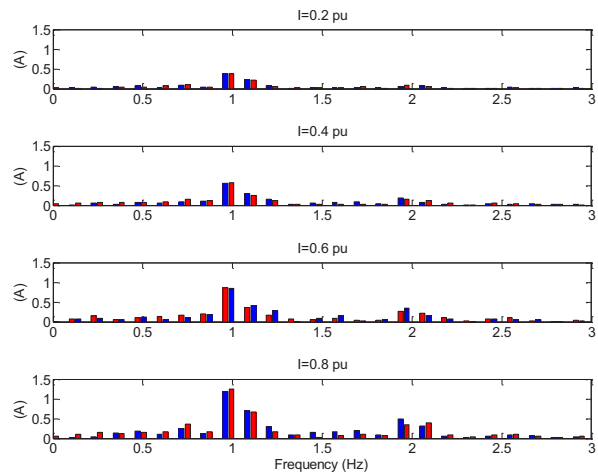

Fig. 13. Frequency spectrum of demodulated $i_{\alpha\beta}$ negative-sequence carrier currents for brushless AC operation with I=0.2-0.8 pu at 10rpm

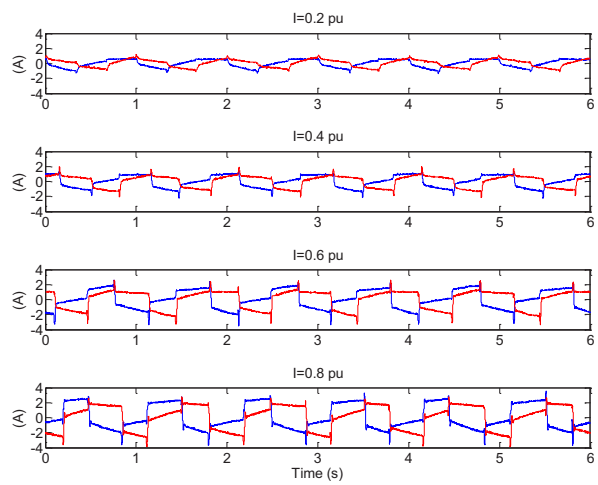

Fig. 14. Demodulated $i_{\alpha\beta}$ negative-sequence carrier currents for brushless DC operation with I=0.2-0.8 pu at 10rpm

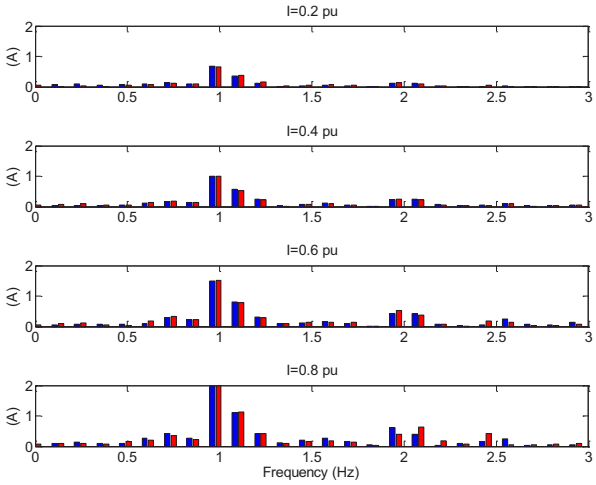

Fig. 15. Frequency spectrum of demodulated $i_{\alpha\beta}$ negative-sequence carrier currents for brushless AC operation with I=0.2-0.8 pu at 10rpm

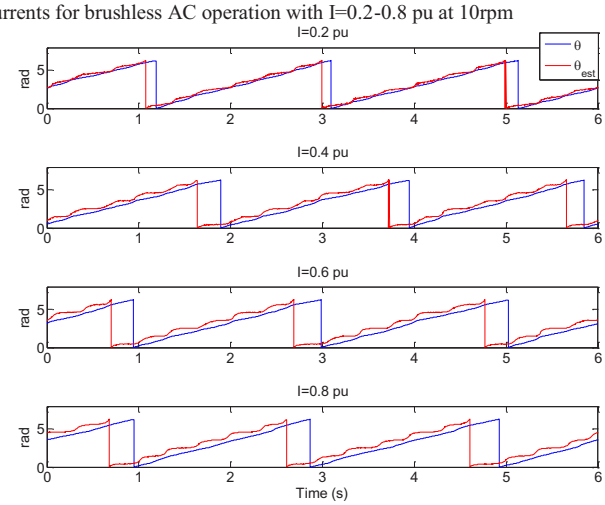

Fig. 16. Measured (blue) and estimated (red) electrical angle at 10rpm for brushless AC operation with I=0.2-0.8 pu at 10rpm

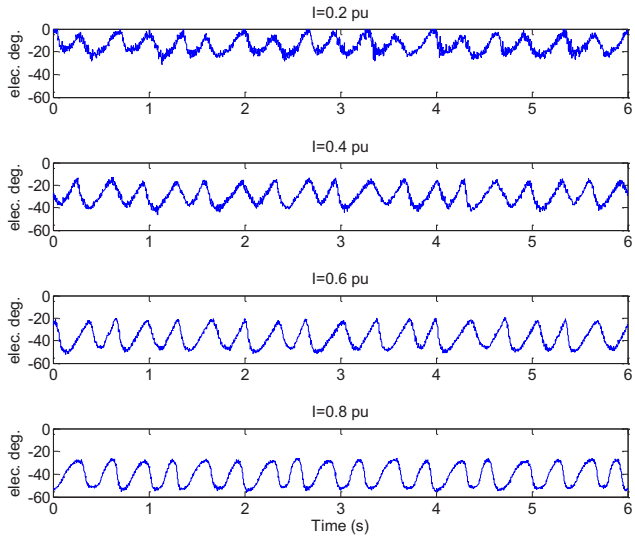

Fig. 17. Electrical angle estimation error at 10rpm for brushless AC operation with I=0.2-0.8 pu at 10rpm

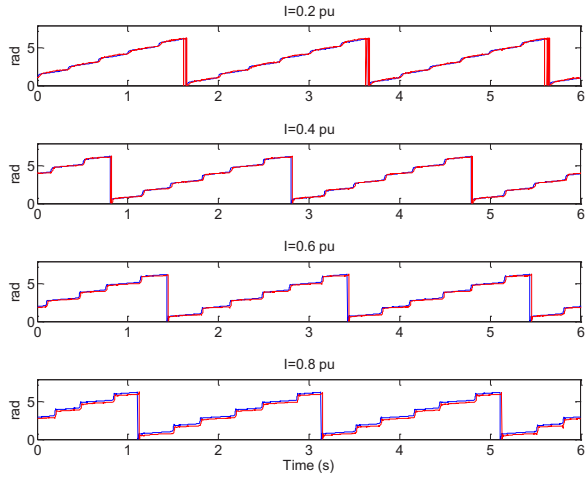

Fig. 18. Measured (blue) and estimated (red) electrical angle at 10rpm for brushless DC operation with I=0.2-0.8 pu.

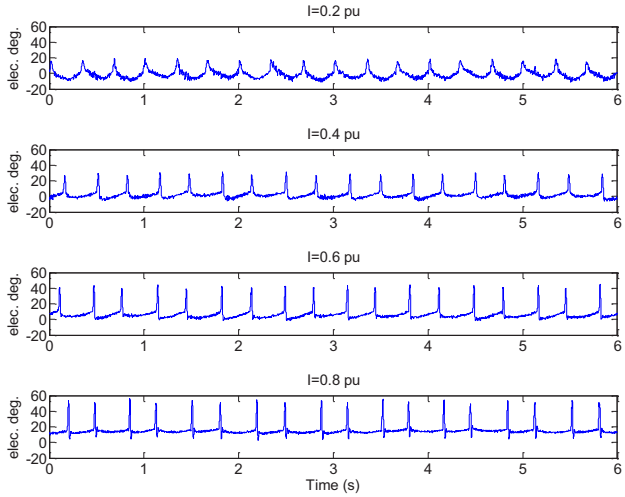

Fig. 19. Electrical angle estimation error at 10rpm for brushless DC operation with I=0.2-0.8 pu.

VI. Conclusions

A finite element analysis based procedure for interior permanent magnet machine design, that incorporates a high-frequency signal injection sensorless position estimation algorithm, has been proposed and exploited in the design of a torque dense machine that exhibits a saliency profile suited to zero- and low-speed sensorless control. The influences of rotor design and operating conditions due to different fundamental current excitations have been investigated by means of detailed finite element analyses and experimental comparisons using the saliency ratio as a measure of the suitability of the machine for sensorless operation. It is shown that, when compared to brushless AC operation, brushless DC operation offers a more gradual reduction in saliency as torque demand increases, suggesting that with DC

excitation sensorless control is possible over a greater torque range. Experimental results have confirmed the potential to extend the range for sensorless operation with DC excitation.

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